

Exponentials and Logarithms 3

Review

1. $y \propto x^2 \Rightarrow y = kx^2$ $0.5 = 4k \Rightarrow k = \frac{1}{8}$

$$\begin{aligned} y &= kx^2 \\ &= \frac{1}{8} \times 5^2 \\ &= \frac{25}{8} \end{aligned}$$

2. $\cos^2 \theta + \cos \theta = \sin^2 \theta$

$$\cos^2 \theta + \cos \theta - 1 + \cos^2 \theta = 0$$

$$2 \cos^2 \theta + \cos \theta - 1 = 0$$

$$(2 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$\cos \theta = \frac{1}{2} \text{ or } \cos \theta = -1$$

$$\text{So } \theta = 60^\circ, 180^\circ, 300^\circ$$

3. d) $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$

4. $\tan \theta = \sqrt{3}$

$$\theta = \frac{1}{3}\pi, -\frac{2}{3}\pi$$

5. $\int_1^2 (x^3 + 1)dx$

$$\begin{aligned} &= \left[\frac{x^4}{4} + x \right]_1^2 \\ &= \left(\frac{2^4}{4} + 2 \right) - \left(\frac{1^4}{4} + 1 \right) \\ &= 6 - \frac{5}{4} \\ &= \frac{19}{4} \end{aligned}$$

Consolidation

$$1. \quad \begin{aligned} 2e^x - 5 - 3e^{-x} &= 0 \\ 2e^{2x} - 5e^x - 3 &= 0 \end{aligned}$$

So

$$\begin{aligned} e^x &\in \left\{-\frac{1}{2}, 3\right\} \\ x &\in \left\{\ln\left(-\frac{1}{2}\right), \ln(3)\right\} \end{aligned}$$

But $\ln\left(-\frac{1}{2}\right)$ is undefined: $e^x > 0$.

So $x = \ln(3)$.

$$2. \quad \log_3(2x + p) = \log_3(4x^2) - \log_3(3)$$

$$\log_3(2x + p) - \log_3\left(4\frac{x^2}{3}\right) = 0$$

$$\log_3\left(\frac{6x + 3p}{4x^2}\right) = 0$$

$$\frac{6x + 3p}{4x^2} = 1$$

$$4x^2 - 6x + 3p = 0$$

Since the equation has only one real root, the discriminant $b^2 - 4ac = 0$.

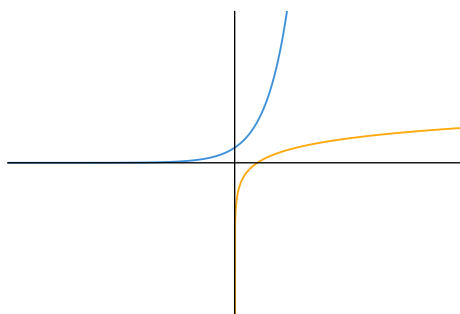
$$(-6)^2 - 4 \times 4 \times 3p = 0$$

$$3p = \frac{36}{16}$$

$$p = \frac{12}{16}$$

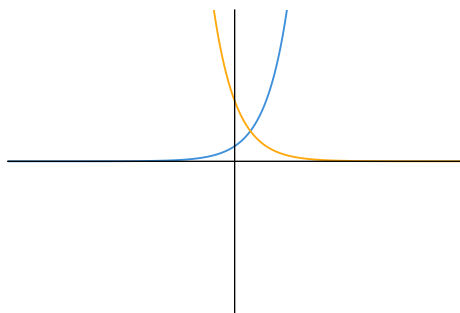
$$= \frac{3}{4}$$

3.



Each curve is the other curve reflected in the line $y = x$.

4. i.



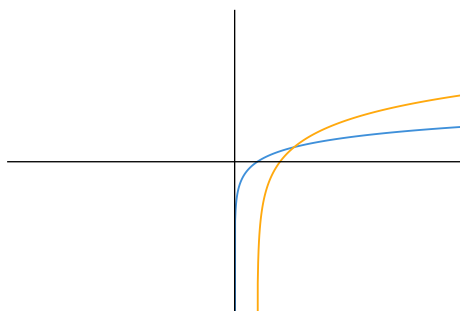
ii.

$$e^x - 4e^{-x} = 0$$

$$e^{x(e^x-4)} = 0$$

But $e^x > 0$ so $x = \ln(4)$ and $y = 4$.

iii.



$$\ln x = \ln(x^2 - 2x + 1)$$

$$x^2 - 3x + 1 = 0$$

$$x \in \left\{ \frac{3 \pm \sqrt{5}}{2} \right\}$$

But $\ln(x-1) > 0$ so $x = \frac{3+\sqrt{5}}{2}$ and $y = \ln\left(\frac{3+\sqrt{5}}{2}\right)$

5. i. If $2x - 1$ is a factor of $f(x) = 2x^3 - x^2 - 8x + 4$, $f\left(\frac{1}{2}\right) = 0$.

$$\begin{aligned} f\left(\frac{1}{2}\right) &= 2\left(\frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 4 \\ &= \frac{1}{4} - \frac{1}{4} - 4 + 4 \\ &= 0 \end{aligned}$$

So therefore $2x - 1$ is a factor of $f(x)$.

ii. Let $z = e^x$. Then

$$2z^3 - z^2 - 8z + 4 = 0$$

$$(2z - 1)(z^2 - 4) = 0$$

$$(2z - 1)(z - 2)(z + 2) = 0$$

So $e^x = \frac{1}{2}$ or $e^x = 2$. $e^x \neq -2$ because $e^x > 0$.

Then

$$x = -\ln(2)$$

or

$$x = \ln(2)$$

Exam Questions

1. a. The student says $2(\log_3 x)^2 = 4 \log_3 x$ when this is false, and the student says $\log_3 8 = 2$, when they most likely are thinking of $\log_2 8 = 3$. $\log_3 8 \neq 2$.

- b. Let $z = \log_3 x$. Then

$$\begin{aligned} 2z^2 - 3z - 2 &= 0 \\ (2z + 1)(z - 2) &= 0 \end{aligned}$$

So $z = 2$ or $z = -\frac{1}{2}$. But $\log_3 x > 0$ so $\log_3 x = 2$. $x = 9$

2. a. Since $f(-1) = 0$, $(x + 1)$ is a factor of $f(x)$. Then

$$\begin{aligned} x^3 + 4x^2 - 7x - 10 &= (x + 1)(x^2 + 3x - 10) \\ &= (x + 1)(x + 5)(x - 2) \end{aligned}$$

- b. From part (a), we get that $x = -1$ or $x = -5$ or $x = 2$. If we let $x = e^y$ such that $x^3 + 4x^2 - 7x - 10 = e^{3y} + 4e^{2y} - 7e^y - 10$, then we get that $e^y \in \{-5, -1, 2\}$. But we know that $e^y > 0$, so $e^y \in \{2\}$.

Therefore, $e^y = 2$ and $y = \ln(2)$.

3. Let $a = e^x$ and $b = e^y$. Then

$$\begin{cases} a - 2b = 3 \\ a^2 - 4b^2 = 33 \end{cases}$$

From equation 1, we get that

$$a = 3 + 2b$$

And therefore

$$\begin{aligned} (3 + 2b)^2 - 4b^2 &= 33 \\ 9 + 12b + 4b^2 - 4b^2 &= 33 \\ 12b &= 24 \\ b &= 2 \end{aligned}$$

Then since $a = 3 + 2b$, $a = 7$.

So

$$\begin{aligned} e^x &= 7 \\ x &= \ln(7) \end{aligned}$$

$$\begin{aligned} e^y &= 2 \\ y &= \ln(2) \end{aligned}$$